

Executive Summary — Surrogate Model Validation Report

Use case: UCFATIGUE_1 | 29 July 2026 | 2 model(s)

REQUIREMENTS PARTIALLY MET

Recommended model: GradientBoosting

Avg R²: 0.9643 — Excellent

Q90 Accuracy — Partially met

Pass: 1g, Vertical maneuver, Vertical gust, Turn, n0

Fail: Frontal gust, Giro

Overfitting check (KS) — Overfitting present

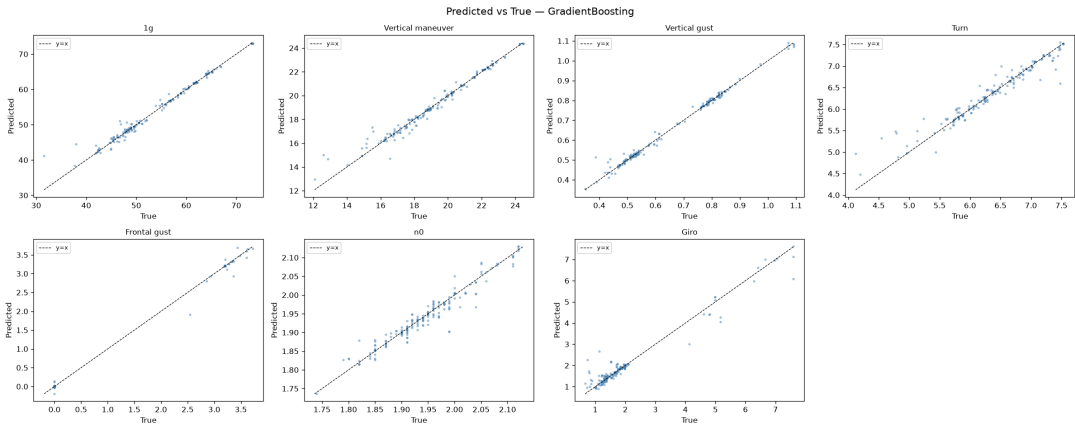
Pass: Vertical maneuver, Frontal gust

Fail: 1g, Vertical gust, Turn, n0, Giro

Project Overview

Use case	UCFATIGUE_1
Inputs	FLAP, Altitude, TAS, Mass, q, gamma, Type_segment, Xcg(%CMA)
Outputs	7: 1g, Vertical maneuver, Vertical gust, Turn, Frontal gust, n0, Giro
Train / Val / Test	80% / 10% / 10%
Accuracy target	Q90 < 10% relative error per output

Predicted vs True — GradientBoosting (test set, up to 5000 points)



Model Performance — GradientBoosting (test set)

Output	R ²	Q90	Target	Gap	Status
1g	0.9797	0.0257	10%	-0.0743	PASS
Vertical maneuver	0.9723	0.0318	10%	-0.0682	PASS
Vertical gust	0.9914	0.0262	10%	-0.0738	PASS
Turn	0.9286	0.0351	10%	-0.0649	PASS
Frontal gust	0.9964	0.2795	10%	+0.1795	FAIL
n0	0.9317	0.0149	10%	-0.0851	PASS
Giro	0.9499	0.2215	10%	+0.1215	FAIL

■ R²≥0.95 ■ R²∈ [0.80, 0.95) ■ R²<0.80 or Q90 fail ■ pass Gap=Q90–target (<0 ⇒ margin)

Analysis & Recommendations

- Significant failures (Frontal gust, Giro): Q90 exceeds target by ≥5%. Investigate inputs and model capacity.

- Overfitting (1g, Vertical gust, Turn, n0, Giro): KS test rejects equal residual distributions.

Overfitting Check (KS Test) — GradientBoosting

Output	KS p-value
1g	0.004
Vertical maneuver	0.161
Vertical gust	0.042
Turn	0.044
Frontal gust	0.974
n0	0.032
Giro	0.017

$p \geq 0.05$ — no overfitting $p < 0.05$ — overfitting detected

Full technical details in the [Technical Review](#) and [Deep Analysis](#).

Technical Review

UCFATIGUE_1

All Models — Full Metrics & Plots

Technical Review — UCFATIGUE_1

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Model Selection

GradientBoosting achieved the highest average R^2 (0.9643), outperforming MLP ($R^2=0.5237$).

Model	Avg R^2	Q90 pass (7)	KS pass (7)
MLP	0.5237	5/7	7/7
GradientBoosting ★	0.9643	5/7	2/7

★ Recommended model. Q90 target: < 10%. KS: $p \geq 0.05$ = no overfitting.

Accuracy Assessment (Q90 & R^2)

GradientBoosting satisfies Q90 on 5/7 outputs. Passing: 1g, Vertical maneuver, Vertical gust, Turn, n0. Failing: Frontal gust, Giro.

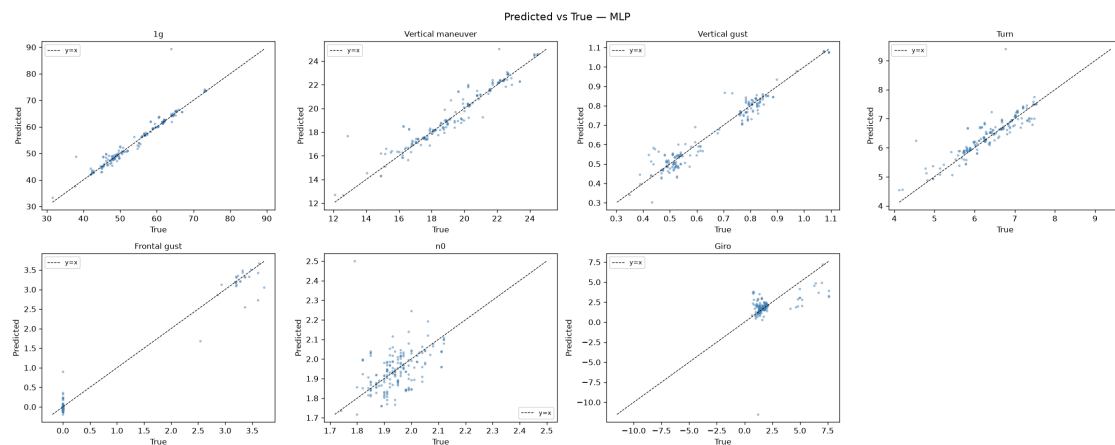
Output	MLP			GradientBoosting ★		
	R^2	Q90	Gap	R^2	Q90	Gap
1g	0.9160	0.0473	-0.0527	0.9797	0.0257	-0.0743
Vertical maneuver	0.9116	0.0554	-0.0446	0.9723	0.0318	-0.0682
Vertical gust	0.9377	0.0918	-0.0082	0.9914	0.0262	-0.0738
Turn	0.7544	0.0783	-0.0217	0.9286	0.0351	-0.0649
Frontal gust	0.9794	11.3300	+11.2300	0.9964	0.2795	+0.1795
n0	-0.7292	0.0723	-0.0277	0.9317	0.0149	-0.0851
Giro	-0.1042	0.9005	+0.8005	0.9499	0.2215	+0.1215

■ $R^2 \geq 0.95$ ■ $R^2 \in [0.80, 0.95)$ ■ $R^2 < 0.80$ or Q90 fail ■ pass Gap=Q90-target (<0 $\Rightarrow$ margin)

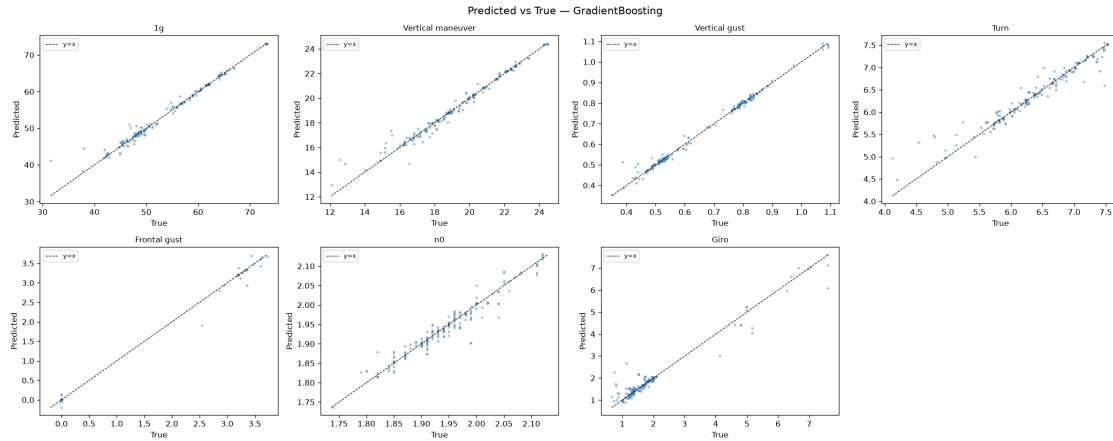
Predicted vs True (Scatter, Test Set)

Each plot shows up to 5 000 test points. Dashed diagonal = perfect prediction.

MLP



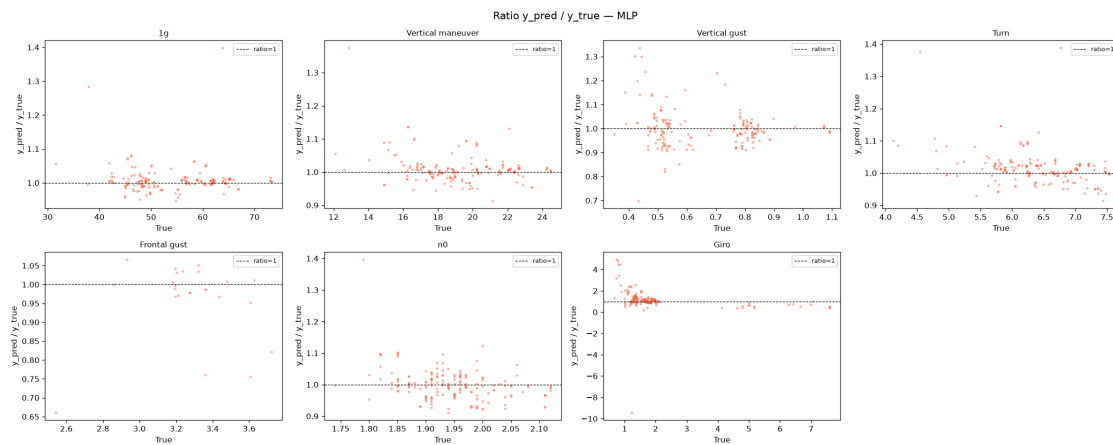
GradientBoosting ★



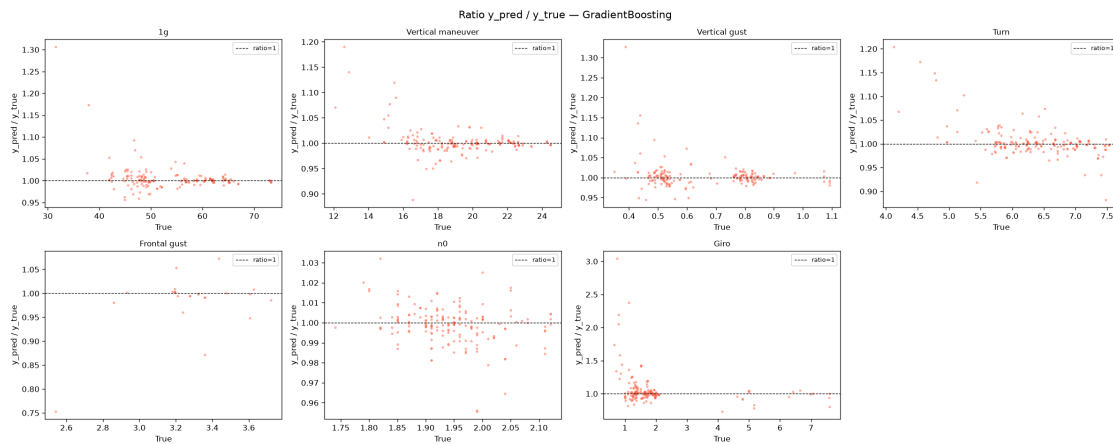
Relative Error Analysis (Ratio Plots)

Ratio $\hat{y}/y$ as a function of the true value. A ratio of 1.0 (dashed) = perfect prediction.

MLP



GradientBoosting ★



Overfitting Assessment (KS Test)

KS test compares residual distributions on train vs. test.

Overfitting detected on: **1g, Vertical gust, Turn, n0, Giro**. Increase regularisation before deployment.

Output	MLP	GradientBoosting
1g	0.473	0.004
Vertical maneuver	0.180	0.161
Vertical gust	0.635	0.042
Turn	0.078	0.044
Frontal gust	0.736	0.974
n0	0.378	0.032
Giro	0.781	0.017

■ $p \geq 0.05$ — no overfitting ■ $p < 0.05$ — overfitting detected

Data Split Quality (VTP Analysis)

VTP confirms unbiased split. No isolated test points; chi-squared passes.

Metric	Value	Pass
Residual voxel proportion (≤ 0.05)	0.000	✓
Valid test proportion	0.931	
Phacking test proportion	0.006	
Isolated test proportion	0.063	
Chi ² p-value (≥ 0.05)	0.7969	✓

Improvement Roadmap

Prioritised actions based on the failure analysis.

- Significant failures (Frontal gust, Giro): Q90 exceeds target by $\geq 5\%$. Investigate inputs and model capacity. Review input features and consider a more expressive architecture.
- Overfitting (1g, Vertical gust, Turn, n0, Giro): KS test rejects equal residual distributions. Increase regularisation or reduce model capacity.

Deep Analysis

UCFATIGUE_1

Winner Model — Full Validation

Deep Analysis — GradientBoosting

Mirrors `validation_template.ipynb` — all computations from validation CSVs

- 3.1 Data Overview
- 3.2 Train-Test Split Analysis
- 3.3 Error Quantification — $P(E)$
- 3.4 $P(E|X)$ — Error Conditional on Inputs
- 3.5 $P(E|Y)$ — Error Conditional on Outputs
- 3.6 Uncertainty Analysis

Data Overview

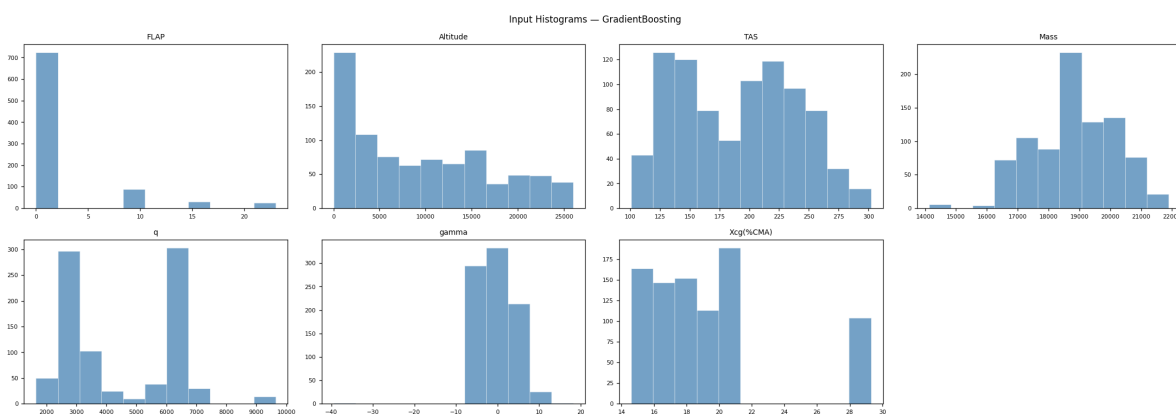
Statistical summary of inputs and outputs for GradientBoosting (all data: train + val + test combined). IQR = interquartile range; Skew and Kurt are the Fisher skewness and excess kurtosis. P10/P50/P90 are the 10th, 50th, and 90th percentiles.

Input Statistics

Input Variable Statistics — GradientBoosting (all data)

Input	N	Mean	Std	IQR	Skew	Kurt	P10	P50	P90
FLAP	869	2.219	5.37	0	2.454	5.311	0	0	10
Altitude	869	9295	7702	1.3e+04	0.507	-0.969	600	8000	2.1e+04
TAS	869	190.9	48.87	86	0.092	-1.138	132	197.6	256
Mass	869	1.887e+04	1325	2230	-0.295	-0.047	1.716e+04	1.89e+04	2.058e+04
q	869	4510	1845	3494	0.339	-1.150	2646	3580	6458
gamma	869	-0.2667	4.573	7.648	-0.523	5.130	-5.873	0	5.64
Xcg(%CMA)	869	19.41	4.095	4.15	1.404	1.221	15.3	18.5	29

Input Histograms

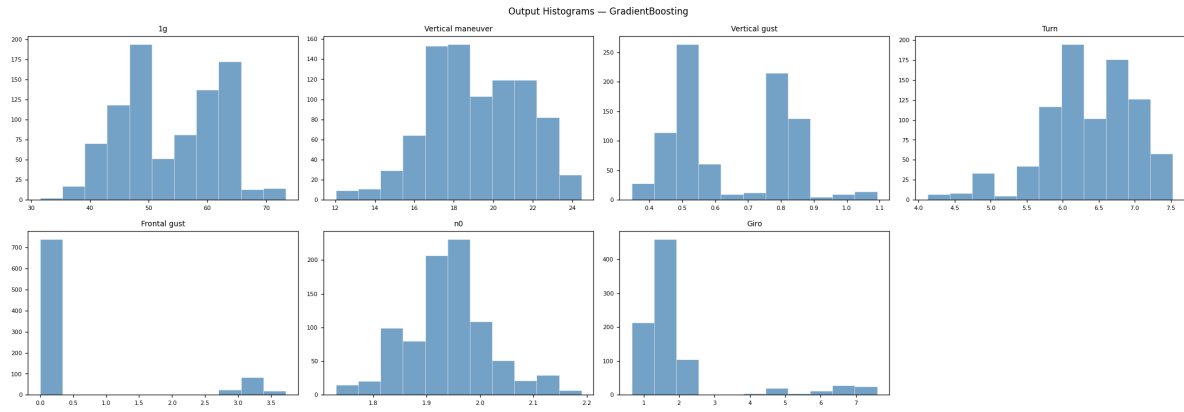


Output Statistics

Output Variable Statistics — GradientBoosting (all data)

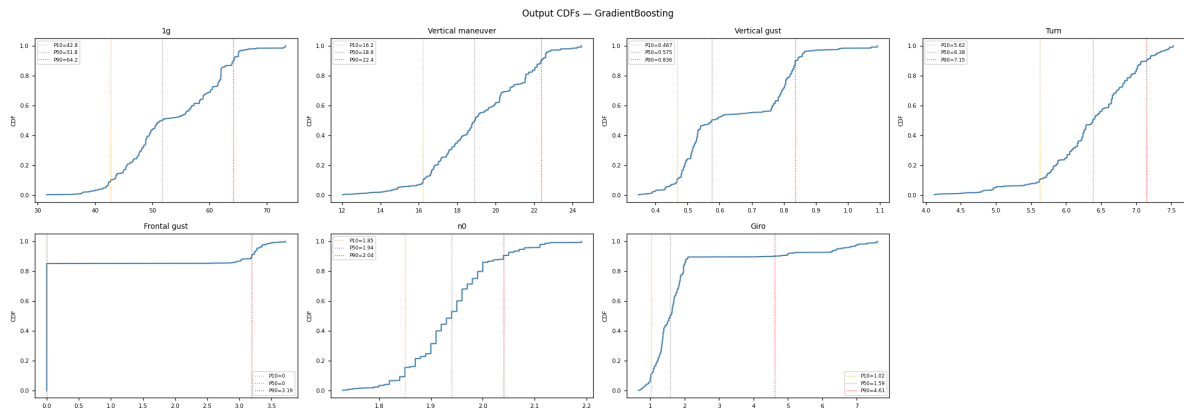
Output	N	Mean	Std	IQR	Skew	Kurt	P10	P50	P90
Ig	869	53.62	8.296	14.19	0.061	-0.965	42.78	51.78	64.19
Vertical maneuver	869	19.14	2.454	4.209	-0.108	-0.480	16.23	18.89	22.36
Vertical gust	869	0.6494	0.1681	0.2963	0.292	-1.132	0.467	0.575	0.836
Turn	869	6.362	0.6415	0.82	-0.679	0.658	5.623	6.382	7.151
Frontal gust	869	0.4806	1.154	0	1.995	2.013	0	0	3.195
n0	869	1.939	0.07768	0.08	0.319	0.538	1.85	1.94	2.04
Giro	869	1.996	1.533	0.575	2.544	5.271	1.025	1.587	4.614

Output Histograms



Output Cumulative Distributions

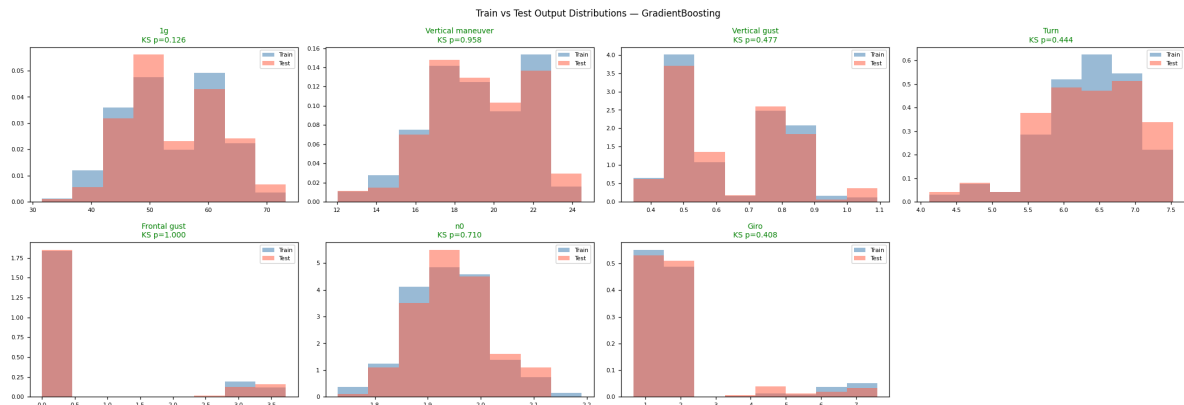
Vertical markers show P10, P50, and P90.



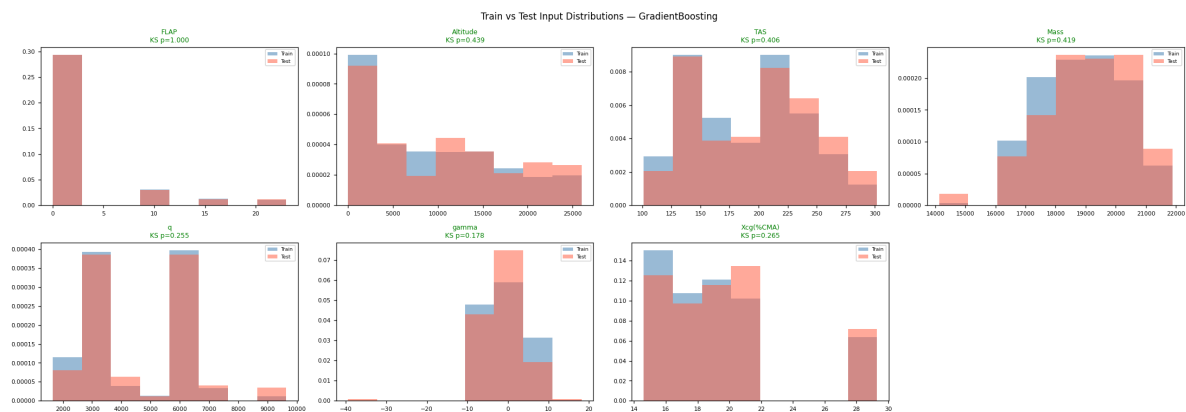
Train-Test Split Analysis

A well-designed split produces training and test distributions that are similar but not identical. The Kolmogorov-Smirnov (KS) test and the Anderson-Darling (AD) test are used to assess distributional similarity. H_0 : both sets come from the same distribution. $p < 0.05$ rejects H_0 (undesirable if too low; undesirable if the split was done by stratification and p is too high).

Output Distributions: Train vs Test



Input Distributions: Train vs Test



KS and AD Test Results

KS and AD Tests: Train vs Test — GradientBoosting

Output	KS p-val	KS	AD p-val	AD
1g	0.1260	Pass	0.0017	Fail
Vertical maneuver	0.9577	Pass	0.0025	Fail
Vertical gust	0.4768	Pass	0.0025	Fail
Turn	0.4441	Pass	0.0015	Fail
Frontal gust	1.0000	Pass	0.0025	Fail
n0	0.7097	Pass	0.0025	Fail
Giro	0.4079	Pass	0.0025	Fail

Error Quantification — P(E)

Two error metrics are studied:

- Residue: $e_i = y_i - \hat{y}_i$. A centred ($\mu \approx 0$), symmetric distribution indicates an unbiased model.
- Absolute Error: $|e_i|$. Always non-negative; its Q90 must satisfy the accuracy requirement.

Residue Statistics Table

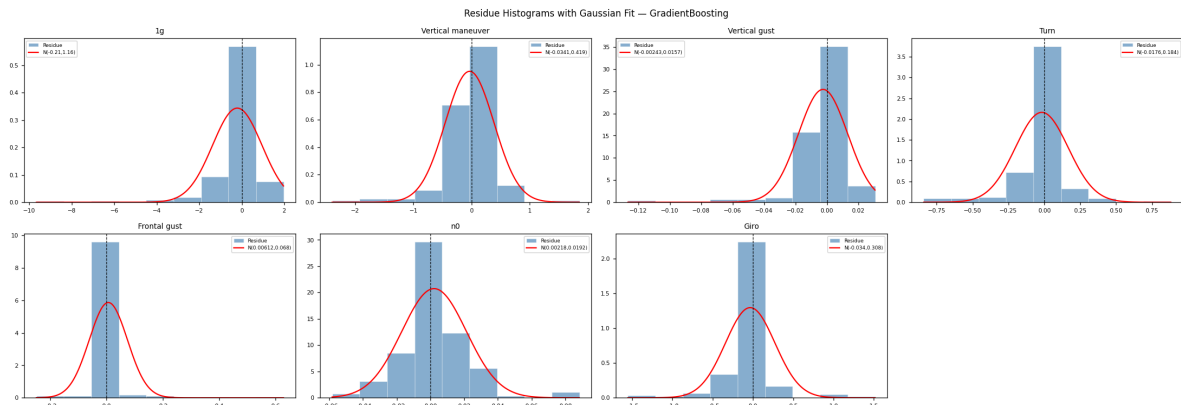
Mean highlighted: green $|\mu|/\sigma < 0.10$, yellow < 0.30 , red ≥ 0.30 .

Residue Statistics ($y - \hat{y}$) — GradientBoosting (test set)

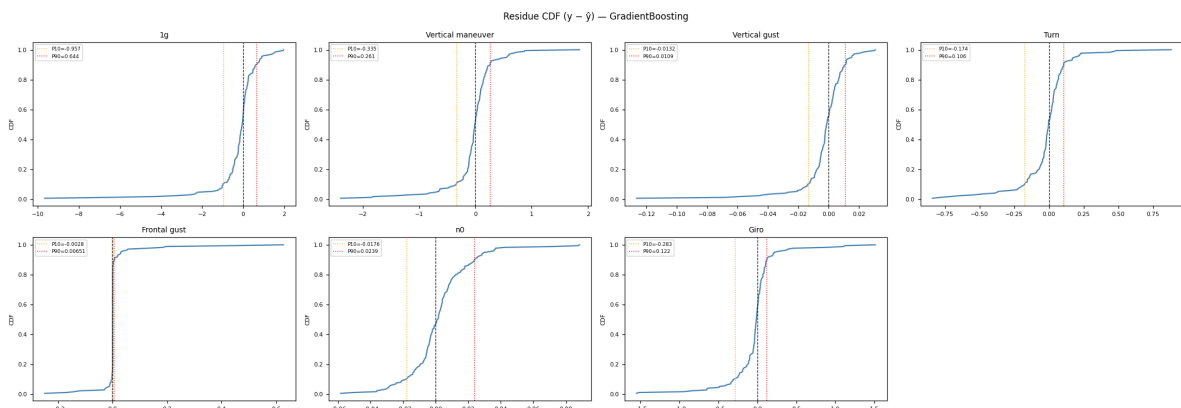
Output	Mean	Std	IQR	Skew	Kurt	P10	P50	P90
1g	-0.2102	1.162	0.5944	-4.361	29.212	-0.957	-0.04563	0.6442
Vertical maneuver	-0.03415	0.4193	0.2242	-1.612	10.804	-0.3346	-0.007409	0.2613
Vertical gust	-0.00243	0.01568	0.009646	-3.638	23.658	-0.0132	-0.001308	0.01092
Turn	-0.0176	0.1845	0.0983	-0.788	8.000	-0.1737	-0.003051	0.1061
Frontal gust	0.00612	0.06798	1.386e-17	5.569	48.302	-0.002795	9.164e-05	0.006509
n0	0.002183	0.01923	0.01504	0.979	4.620	-0.01758	0.001351	0.02392
Giro	-0.03397	0.3076	0.1418	-0.253	11.125	-0.2828	-0.01135	0.1216

Residue Histograms with Gaussian Fit

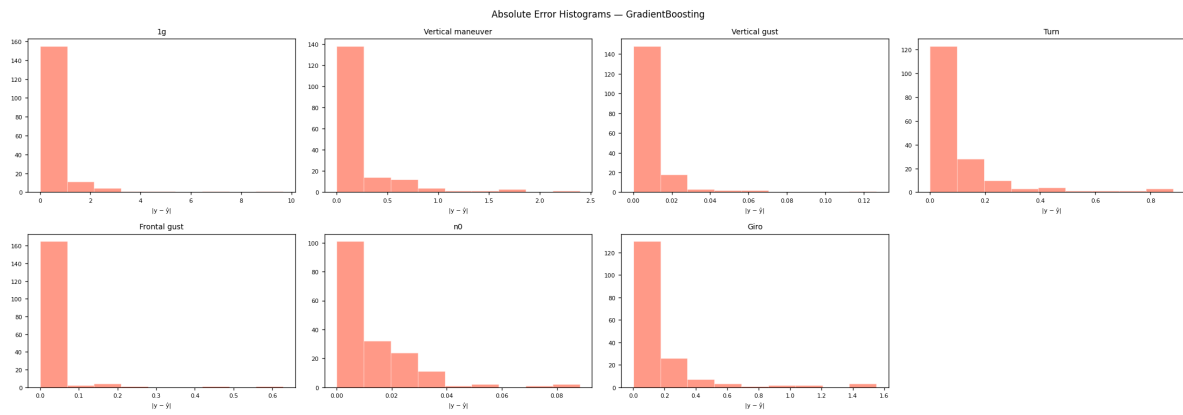
Red curve = fitted normal distribution. Deviation from the normal indicates heavy tails or skew.



Residue CDF

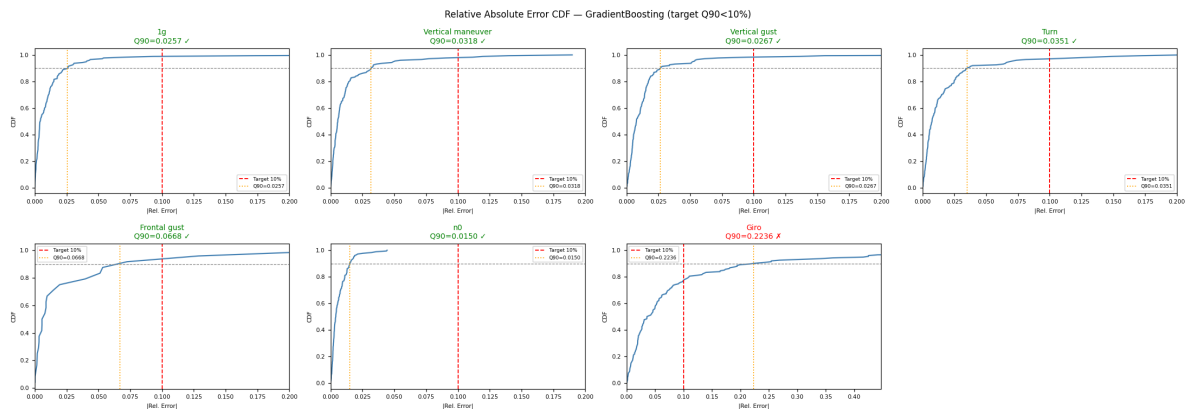


Absolute Error Histograms



Relative Absolute Error CDF — Q90 Accuracy Requirement

Orange dotted = observed Q90; red dashed = target threshold. Green title = PASS; red title = FAIL.

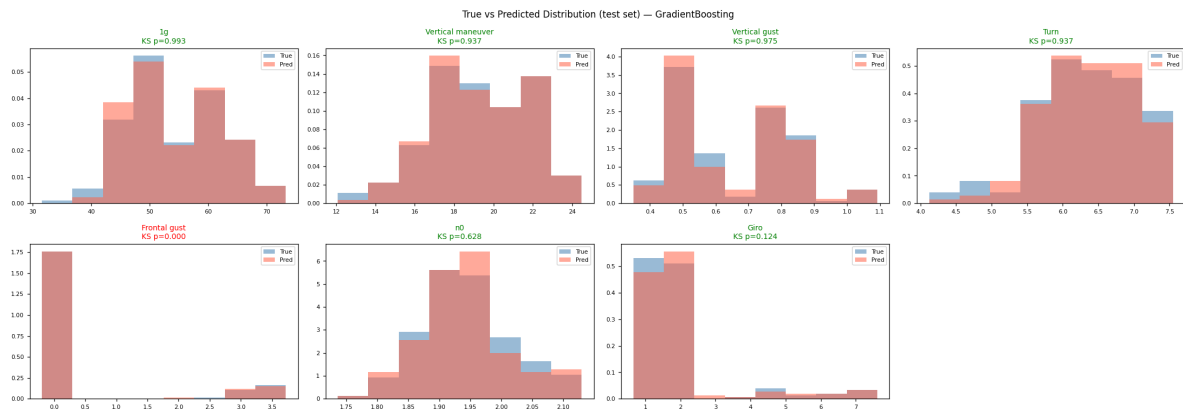


Output	Q90	Target	Status
1g	0.0257	10%	PASS
Vertical maneuver	0.0318	10%	PASS
Vertical gust	0.0267	10%	PASS
Turn	0.0351	10%	PASS
Frontal gust	0.0668	10%	PASS
n0	0.0150	10%	PASS
Giro	0.2236	10%	FAIL

■ $p \geq 0.05$ — pass ■ $p < 0.05$ — fail

True vs Predicted Distribution Comparison

KS test comparing the distribution of true vs. predicted values on the test set. A significant difference indicates the model is not reproducing the output distribution faithfully.



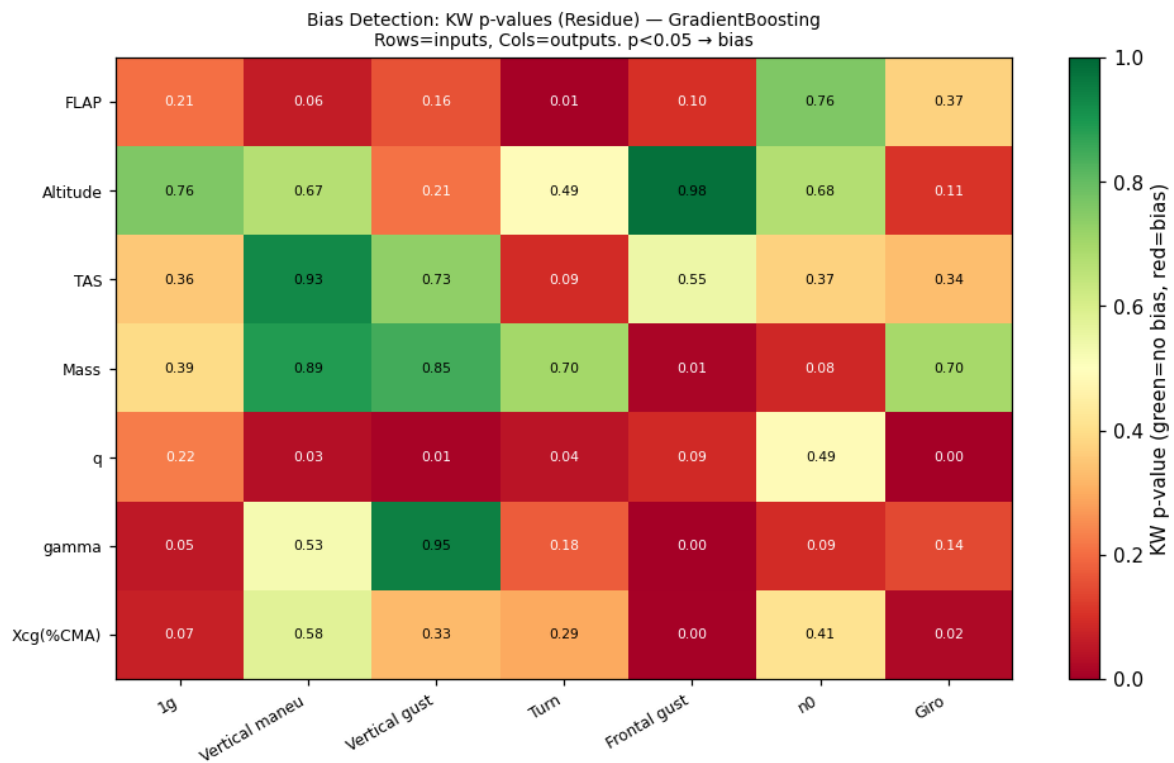
P(E|X) — Error Conditional on Inputs

We study whether the model error depends on the input variables. Two approaches are used: (1) a Kruskal-Wallis (KW) test on binned inputs (Sturges rule for bin count), and (2) violin plots showing the error distribution within each input bin.

Significant input-conditional bias ($p < 0.05$) detected for: **FLAP**→Turn, **Mass**→Frontal gust, **q**→Vertical maneuver, **q**→Vertical gust, **q**→Turn, **q**→Giro, **gamma**→Frontal gust, **Xcg(%CMA)**→Frontal gust, **Xcg(%CMA)**→Giro. These inputs explain residual variance — consider polynomial features or interaction terms.

Bias Detection Heatmap (KW p-values)

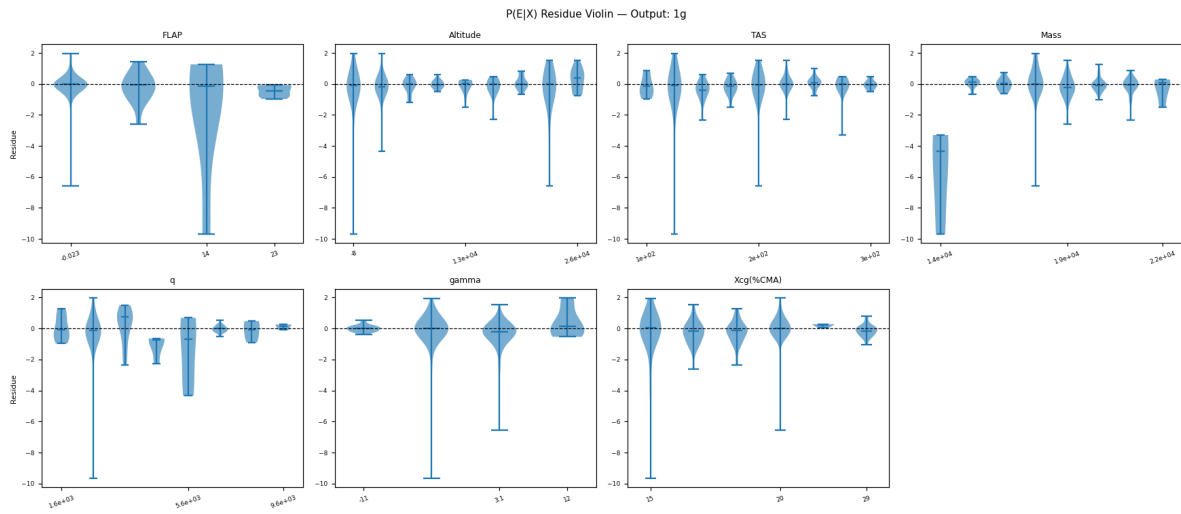
Rows = inputs, columns = outputs. Green ($p \geq 0.05$) = residue uniform across input bins; red ($p < 0.05$) = significant bias.



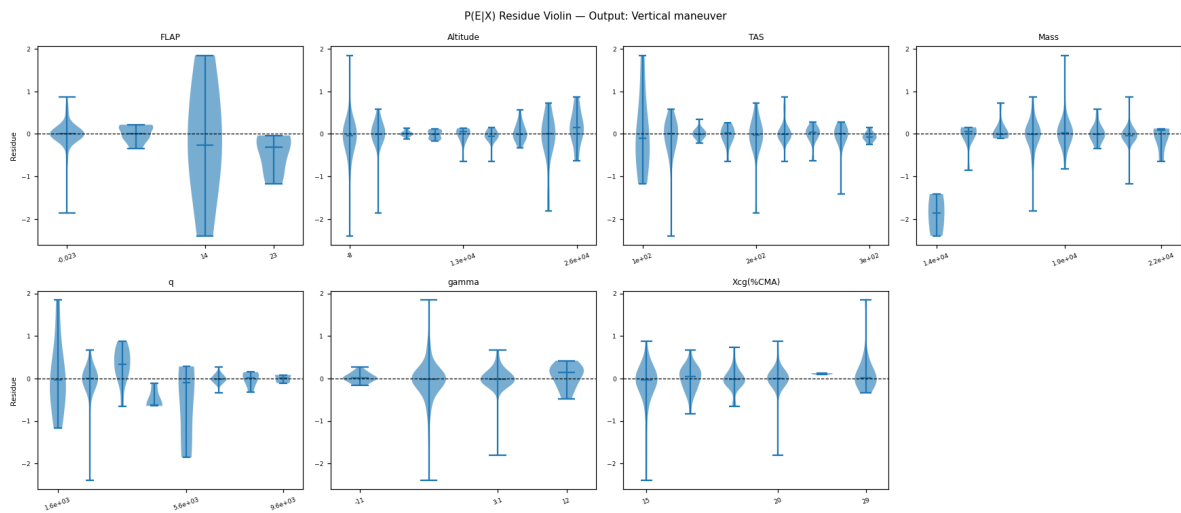
Residue Violin Plots per Output (binned by input)

Each panel shows the residue distribution for each input bin. A flat median line (dashed at 0) and symmetric violins indicate absence of bias.

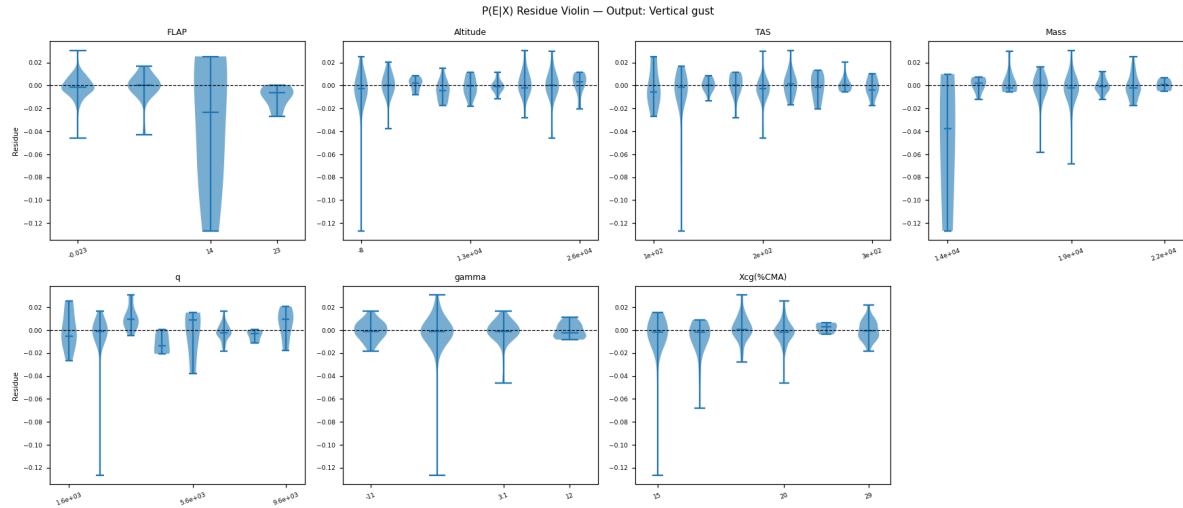
Residue — 1g



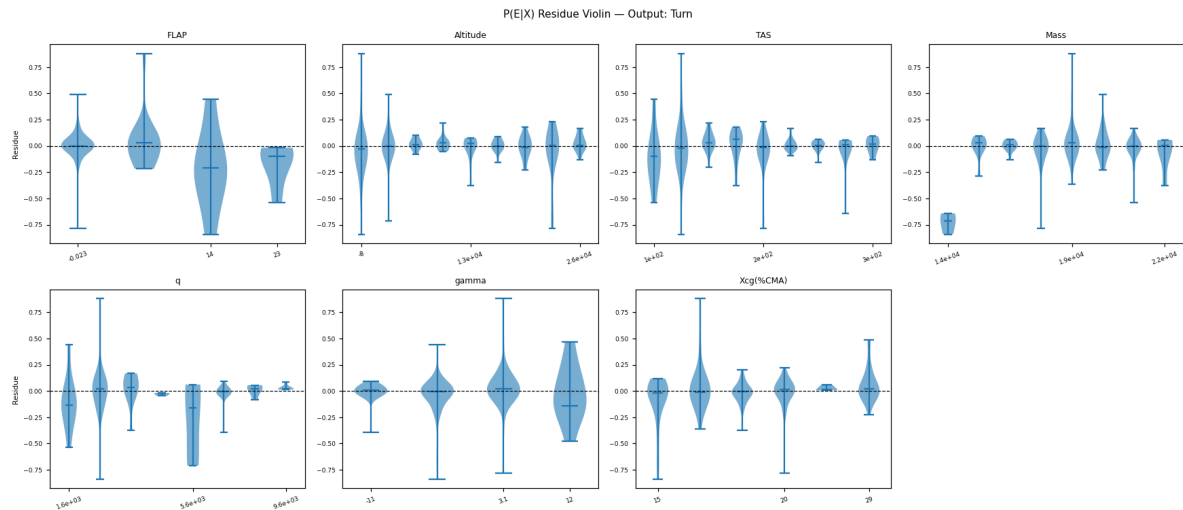
Residue — Vertical maneuver



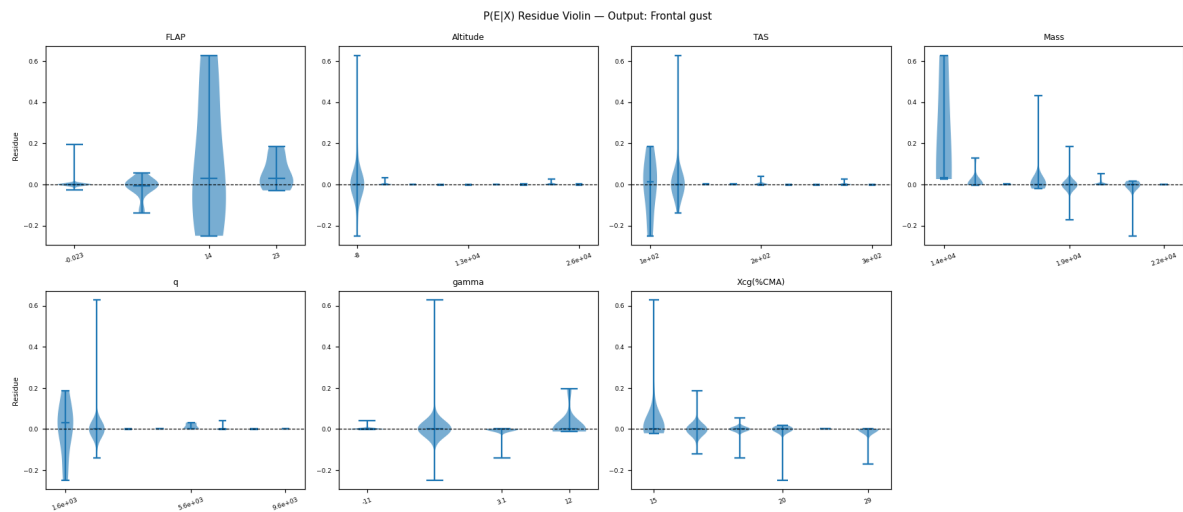
Residue — Vertical gust



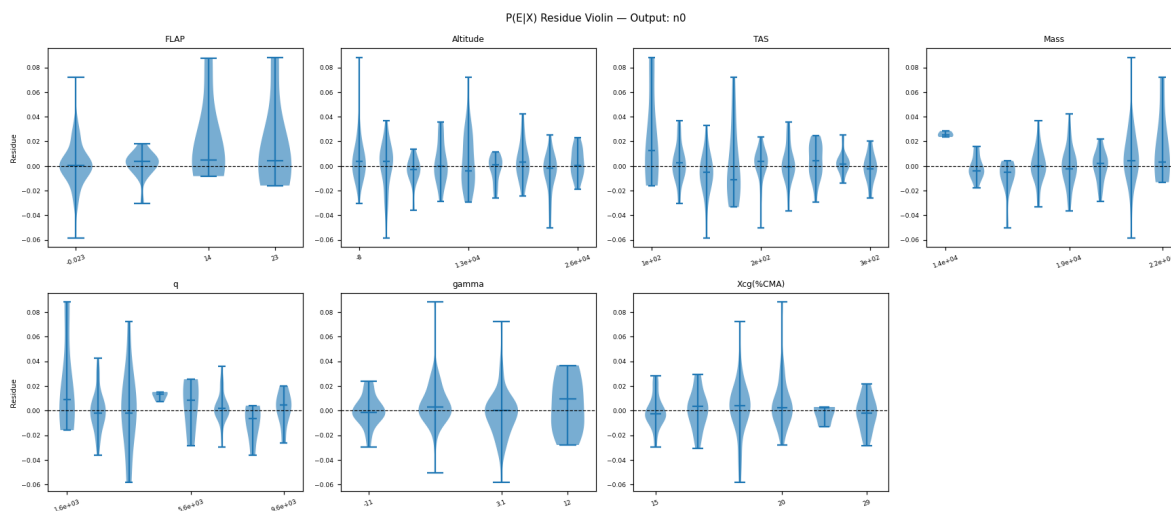
Residue — Turn



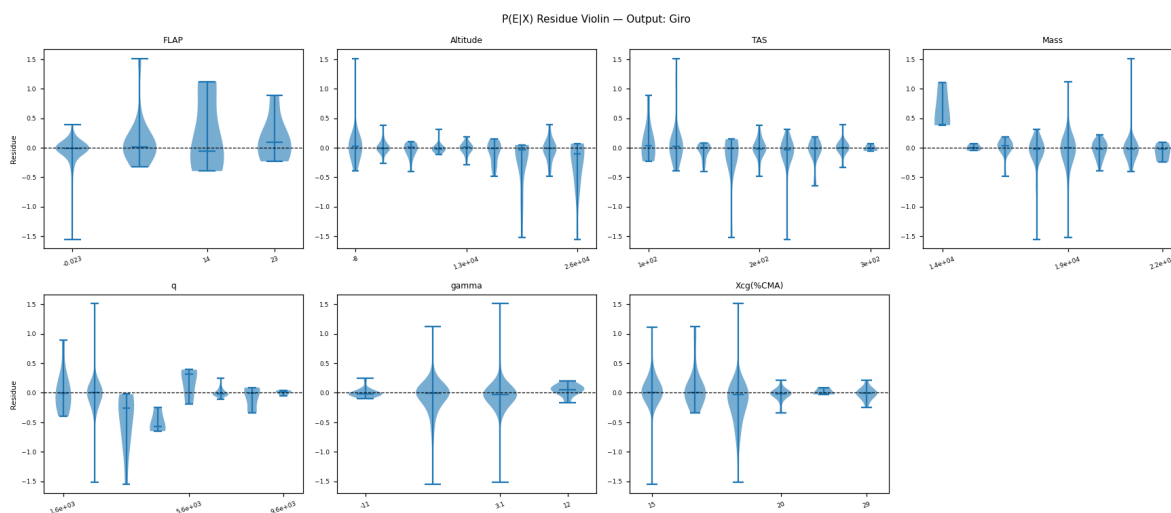
Residue — Frontal gust



Residue — n0

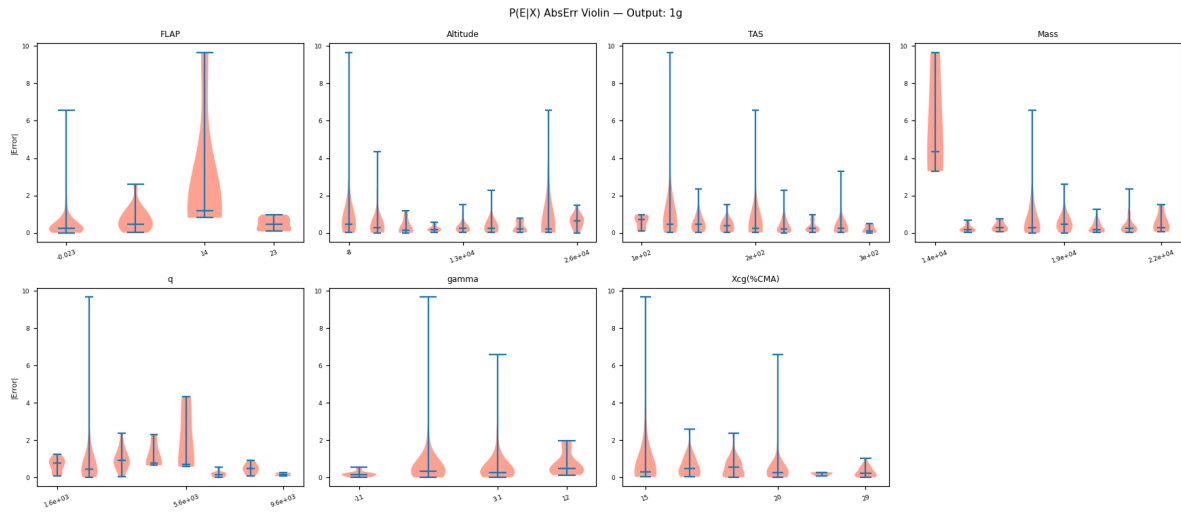


Residue — Giro

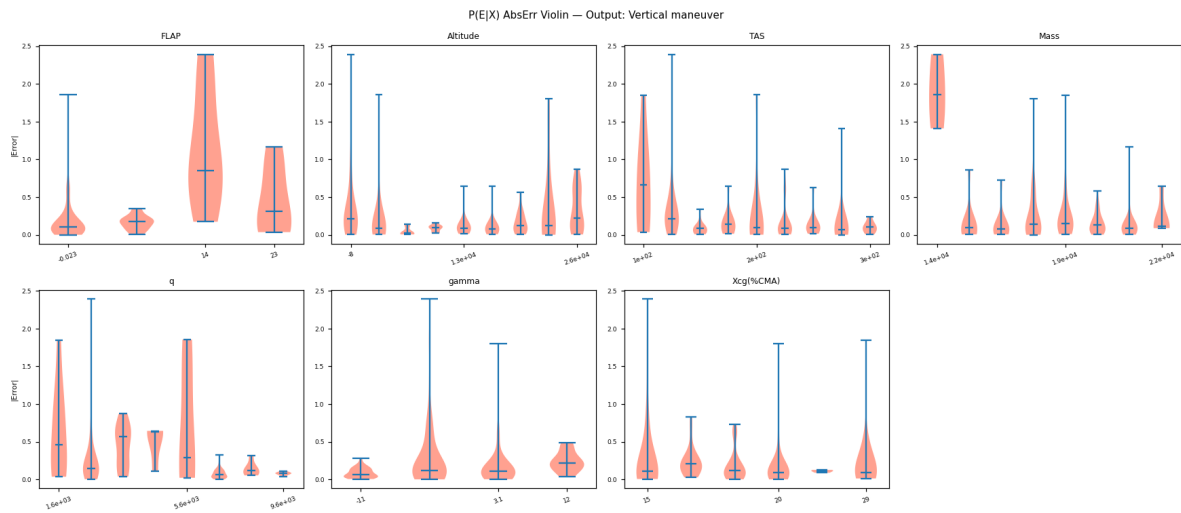


Absolute Error Violin Plots per Output (binned by input)

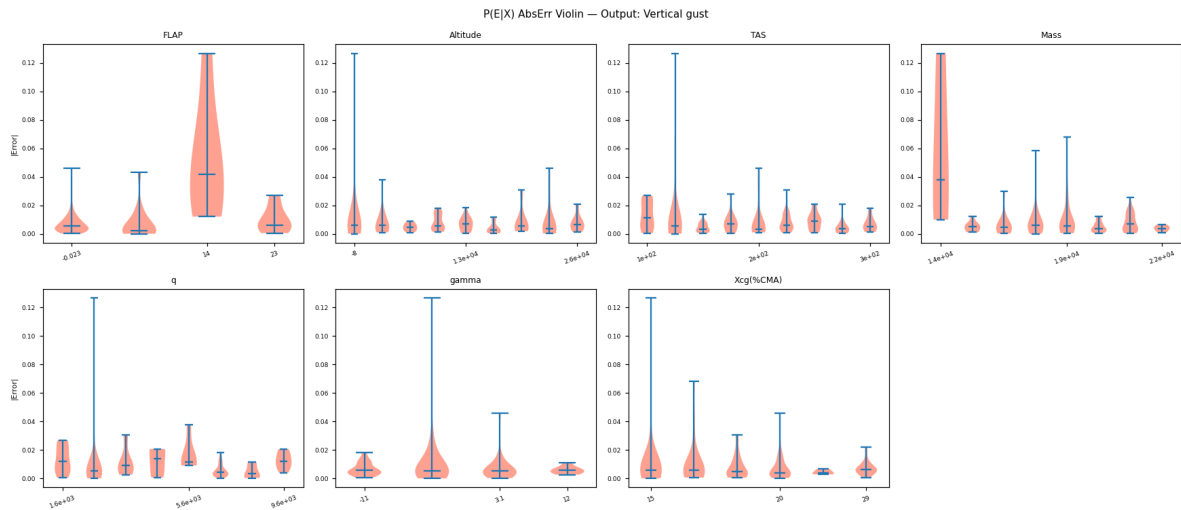
Absolute Error — 1g



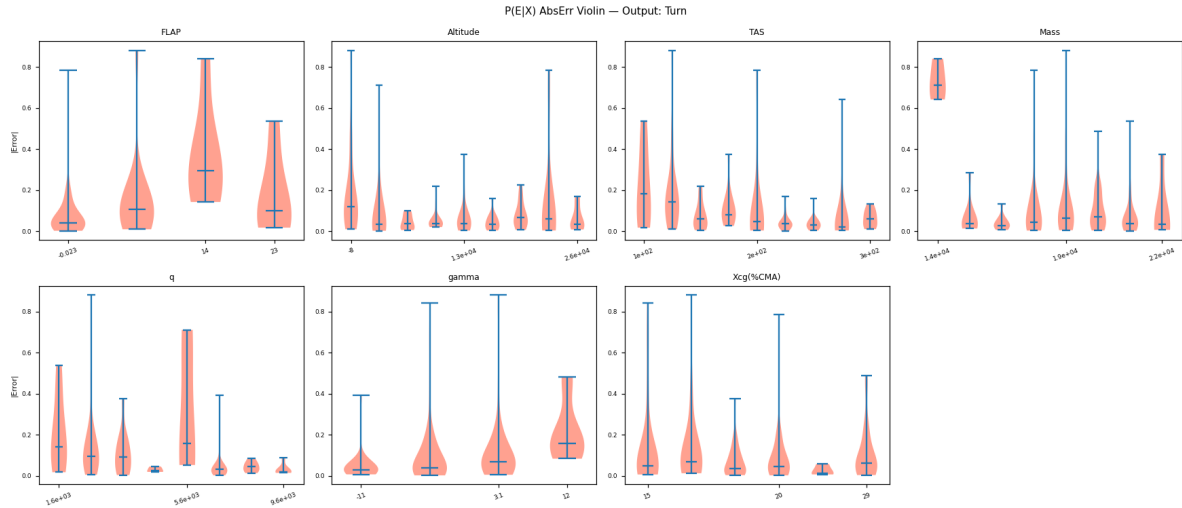
Absolute Error — Vertical maneuver



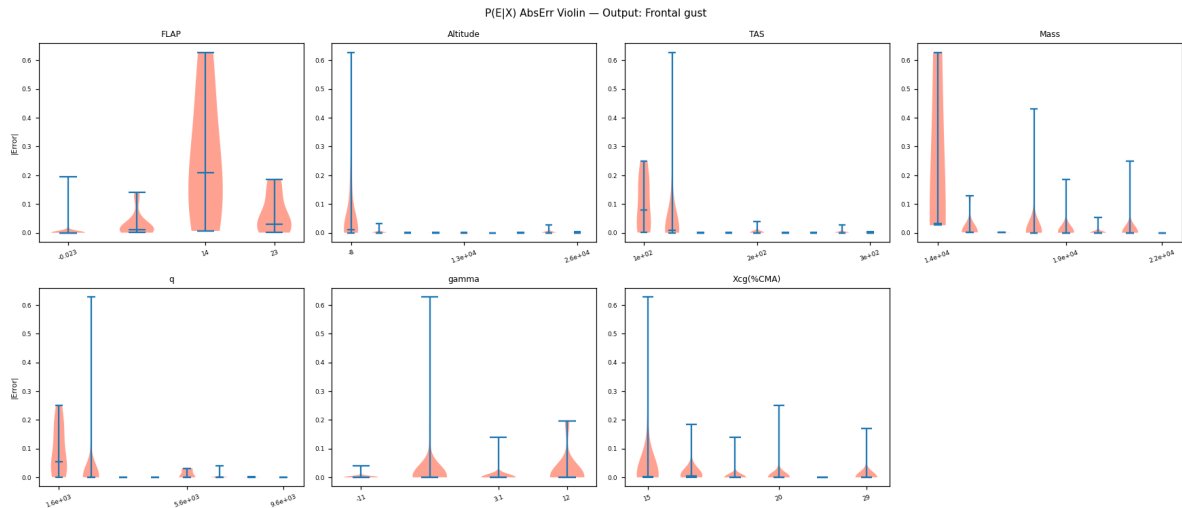
Absolute Error — Vertical gust



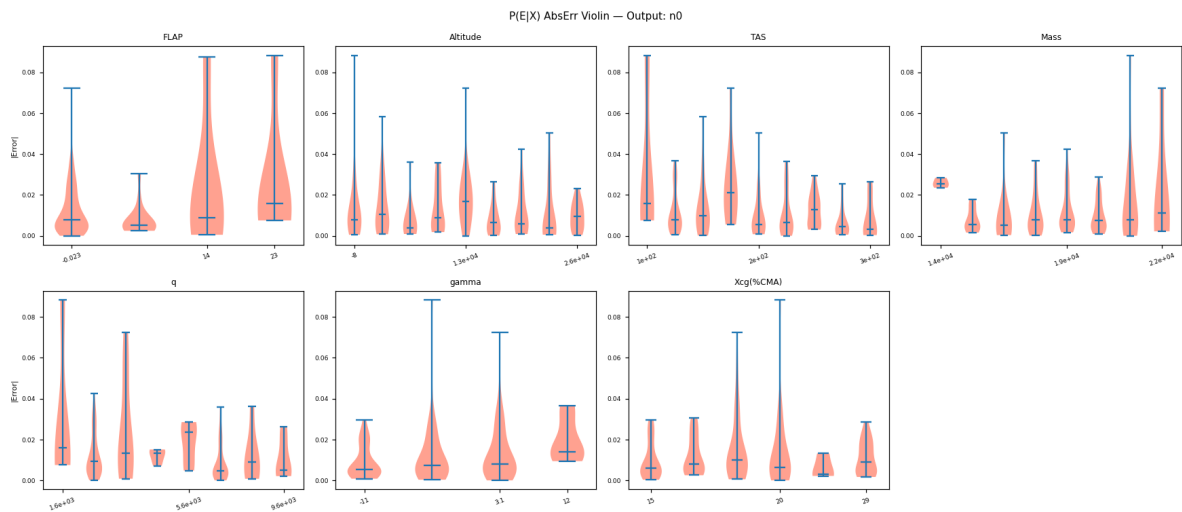
Absolute Error — Turn



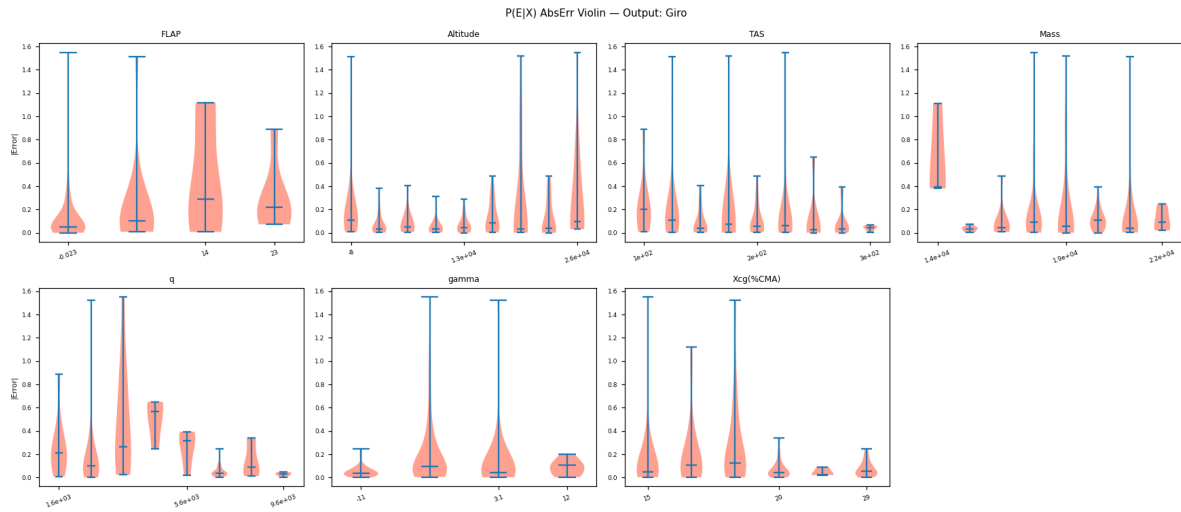
Absolute Error — Frontal gust



Absolute Error — n0



Absolute Error — Giro

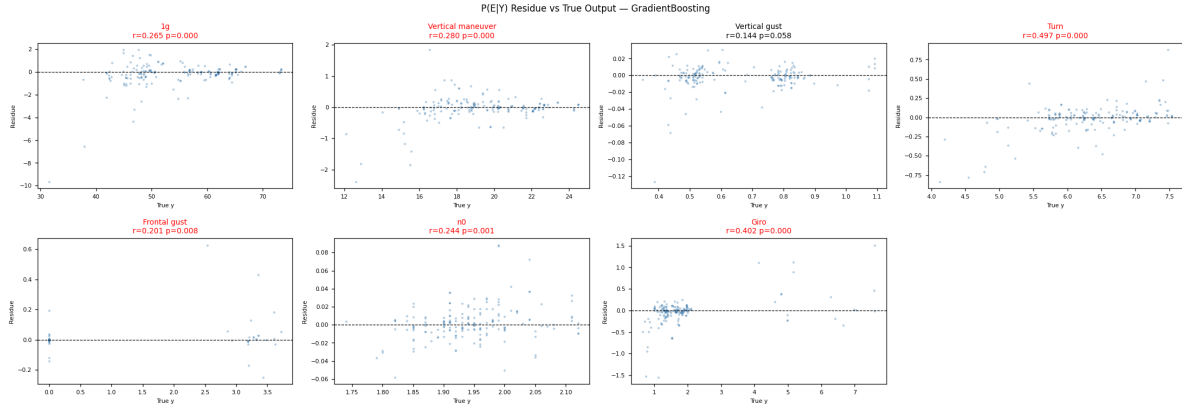


P(E|Y) — Error Conditional on Outputs

We study whether the error depends on the output magnitude. A significant trend (Pearson or Spearman $p < 0.05$) means the model is more accurate in some output ranges than others — a form of heteroscedasticity.

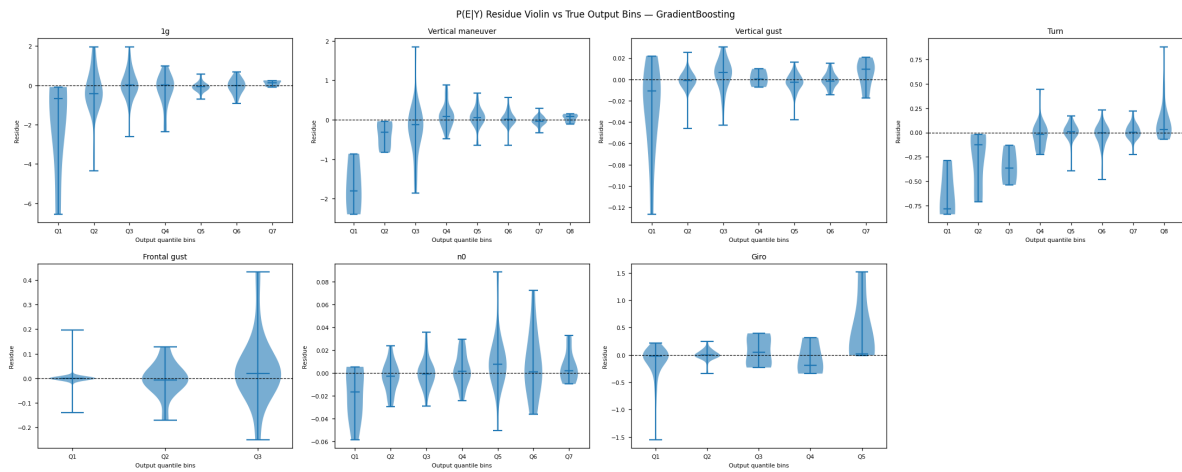
Residue vs True Output — Scatter

Pearson r and p-value shown in title. Red title = significant linear trend.



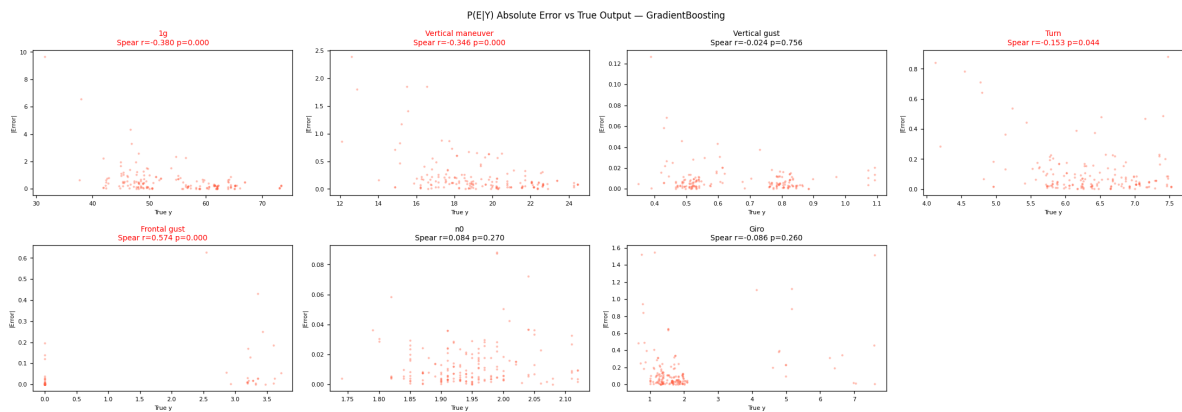
Residue Violin vs True Output Bins

Bins determined by Sturges rule. Median line dashed at 0.

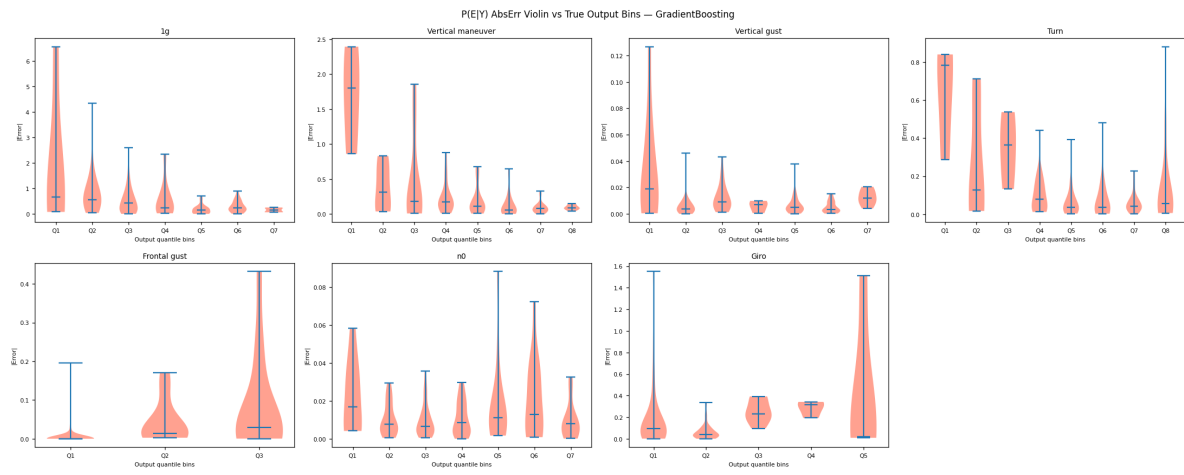


Absolute Error vs True Output — Scatter

Spearman r shown. Red title = significant monotonic trend.



Absolute Error Violin vs True Output Bins



Uncertainty Analysis

A global Gaussian uncertainty model is estimated from the residue standard deviation on the test set. Coverage is the empirical proportion of test points within $\pm k\sigma$ bounds. For a perfectly Gaussian error, $\pm 1\sigma$ should cover 68.3%, $\pm 2\sigma$ 95.4%, $\pm 3\sigma$ 99.7%. Green = within 3%, yellow = within 10%, red = significant deviation.

Sigma Coverage Table

Uncertainty — Sigma Coverage — GradientBoosting
Expected for Gaussian: $\pm 1\sigma=68.3\%$, $\pm 2\sigma=95.4\%$, $\pm 3\sigma=99.7\%$
Green=within 3%, Yellow=within 10%, Red=deviates significantly

Output	σ	Cov $\pm 1\sigma$	Cov $\pm 2\sigma$	Cov $\pm 3\sigma$	Δ from 68.3%	Δ from 95.4%	Δ from 99.7%
1g	1.162	89.7%	96.6%	98.3%	0.214	0.012	0.014
Vertical maneuver	0.4193	85.1%	94.8%	97.1%	0.168	0.006	0.026
Vertical gust	0.01568	87.9%	96.6%	98.3%	0.196	0.012	0.014
Turn	0.1845	85.6%	93.1%	97.1%	0.173	0.023	0.026
Frontal gust	0.06798	94.8%	96.0%	98.3%	0.265	0.006	0.014
n0	0.01923	76.4%	96.6%	97.7%	0.081	0.012	0.020
Giro	0.3076	85.6%	93.7%	96.6%	0.173	0.017	0.031

Uncertainty Bounds $\pm 2\sigma$ on Predicted vs True

Orange ribbon = $\pm 2\sigma$ around the prediction. Points outside the ribbon are larger errors than expected under the Gaussian model.

